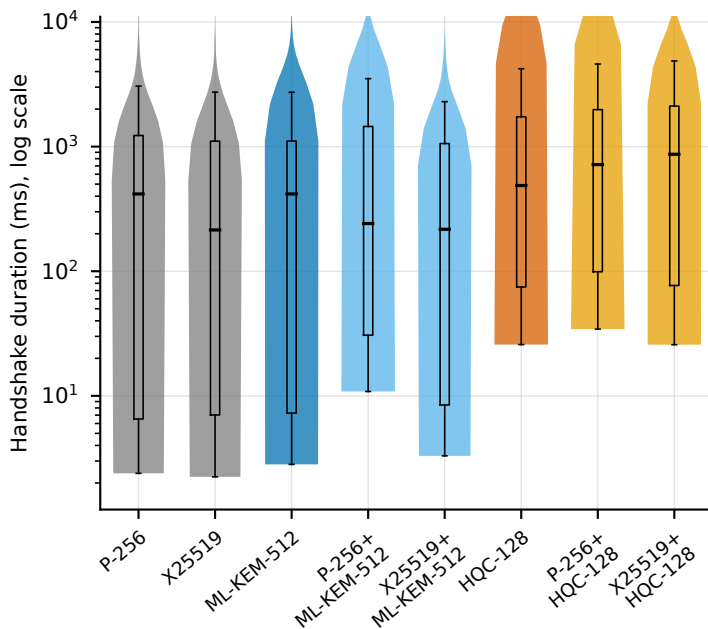


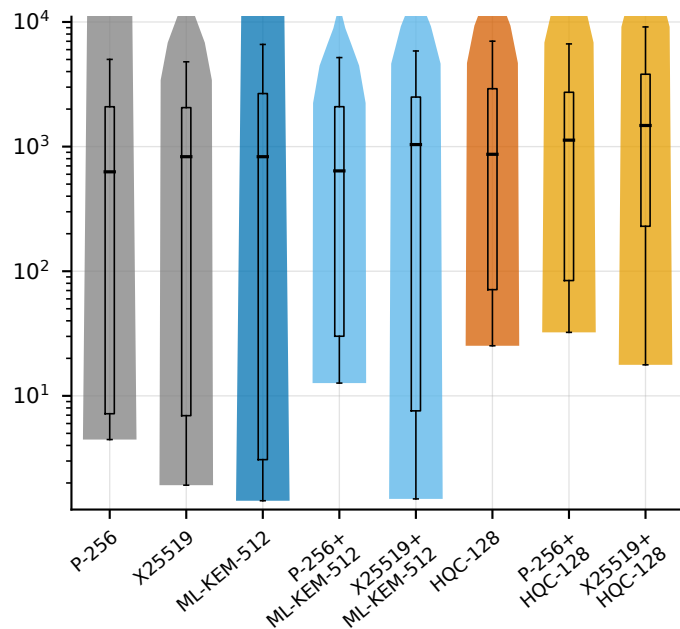
(a)

ed25519 (classical)



(b)

mldsa44 (post-quantum)



(c)

 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)
P-256	1.41k	294.4	52.57k	7.75k	890.2	1.35M
X25519	1.31k	287.1	52.56k	3.81k	557.7	338.03k
ML-KEM-512	1.67k	441.1	108.53k	8.28k	1203.4	2.19M
P-256+ML-KEM-512	2.39k	597.0	216.81k	3.41k	403.5	220.25k
X25519+ML-KEM-512	1.15k	335.9	69.16k	3.58k	650.8	457.60k
HQC-128	5.94k	546.2	460.30k	6.06k	460.2	461.10k
P-256+HQC-128	4.52k	685.0	647.82k	6.37k	595.5	672.82k
X25519+HQC-128	3.39k	427.5	215.61k	7.54k	619.7	901.99k